

The effects of different levels of disfluency on learning outcomes and cognitive load

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Abstract In some cases difficult-to-read learning materials lead to better performance than easy-to-read materials. This phenomenon is called the disfluency effect and is attributed to a subjective feeling of task difficulty resulting in a deeper learning approach. In two studies ($n_1 = 67$, $n_2 = 73$) we investigated the impact of three increasing disfluency levels compared to a fluent condition on learning outcomes, in the second study we additionally addressed effects on cognitive load and time on task. The first study revealed significant differences of the experimental groups and on a descriptive level a reversed u-shape pattern, i.e., learning performance increases with increasing disfluency but decreases when the text becomes too illegible. The same pattern could be found significantly in the second study for recall and transfer performance, engagement and time on task and the reversed pattern for extraneous load. Results indicate that there is in fact an optimal level of disfluency on performance that comes along with increased engagement and that there is a breaking point of disfluency where the extraneous load seems to be too high.

Keywords Disfluency effect · Increasing disfluency levels · Cognitive load · Metacognition · Reversed u-shaped pattern

Introduction

Imagine reading a text in a very difficult to read font. Most educators and instructional researchers might argue that this will impair the learning process, but the concept of desirable difficulties (Bjork 1994) indicates that making learning material harder to learn leads to better performance. Disfluency like the hard to read font is one special kind of these desirable difficulties and is defined as a metacognitive cue of perceived task difficulty

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that prompts more elaborated processing (Alter et al. 2007; Yue et al. 2013). In the last years many studies addressed the disfluency effect and some of them confirmed it, others showed either null effects or even reverse effects (for an overview see Kühl and Eitel 2016). As the positive impact of disfluent material seems to depend on the intensity of disfluency itself (Oppenheimer and Alter 2014), one possible explanation for the different evidence of disfluency might be attributed to different levels of disfluency induced in the studies. In other words low levels of disfluency might be too easy to read whereas with increasing levels of disfluency there might be a point at which the learning material is illegible and actually interferes with learning. Based on this assumption, Diemand-Yauman et al. (2011) suggested that the influence of disfluency on performance follows a reversed u-shaped curve. Based on this, the present research investigates whether learning material with different disfluency levels actually result in a reversed u-shaped pattern on the learning performance.

Disfluency as desirable difficulty

Introducing difficulties in the learning process can improve long-term learning and retention (Bjork 1994). But such difficulties are only desirable, if the challenging character of the chosen method stimulates further cognitive engagement and deeper processing in terms of analytic and elaborative strategies which foster encoding and support learning or long-term performance (Alter et al. 2007; Bjork 1994; Craik and Tulving 1975). Very often, desirable difficulties are provided prior to the learning phase, e.g. by eliciting problem solving, exploratory activities or even productive failure (Kapur 2008). The idea behind these opening activities is to activate learners' prior knowledge, to make them aware of knowledge gaps or to elicit a deep structure emphasizing approach (Kalyuga and Singh 2015; Schwartz and Bransford 1998). There are a variety of desirable difficulties that manipulate the objective difficulty as well as the subjective difficulty (Diemand-Yauman et al. 2011). One way to manipulate the subjective or perceived difficulty only, is through disfluency, which can be achieved by using hard to read fonts or faded font colors. The disfluently presented information may also act as a desirable opening trigger for a deeper learning approach as it may act as a metacognitive cue that influences the learner's subsequent cognitive processing. This metacognitive regulation mechanism is explained in the disfluency theory of Alter et al. (2007). He refers to dual processing theories (James 1950; Tverski and Kahneman 1974), according to which there are two different reasoning systems: System 1 works intuitively, effortlessly, quickly and associatively; System 2 is conscious, effortful, slow and analytic. The activation of System 1 and System 2 depends on the subjective or perceived difficulty of the cognitive task (Alter et al. 2007). With disfluent learning material learners perceive the task as more difficult and metacognitively regulate their learning approach by activating System 2. Fluent learning material on the other hand will be perceived as rather easy and learners should be more likely to only activate System 1. Hence a cognitive task can range from a subjective perception from effortless to highly effortful and is associated with a metacognitive experience that varies from fluent to disfluent (Alter and Oppenheimer 2009). Hence, due to the disfluency theory, disfluency does not only require learners to take a closer or longer look on the difficult to read text but affect their learning approach in general. With this, it is possible to explain that disfluency can transfer to improved learning outcomes on all levels of processing and not only to better scores that would link to intensified word processing like word recognition or recall tasks.

Alter et al. (2007) demonstrated in several studies that fluency in fact leads to a more superficial and heuristic learning—resulting in less subsequent correct answers—in contrast to disfluency that seems to enable a more elaborated and systematic processing. Song and Schwarz (2008) as well as Corley et al. (2007) provided further evidence for a more careful and comprehension facilitating learning through disfluent material. Beyond that, other studies also showed a positive effect of disfluent learning material on performance. Sungkhasettee et al. (2011) showed that inverted words led to better recall performances compared to upright ones. Furthermore, Diemand-Yauman et al. (2011) found the disfluency effect on retention not only in an experimental setting but also in a real learning context. Eitel et al. (2014) could also demonstrate the disfluency effect in at least one study. But there are also studies which did not find a positive effect of disfluency but rather a neutral or partly even a negative effect. Rhodes and Castel (2008) investigated the influence of font sizes, but found no performance differences. Yue et al. (2013) as well as Eitel et al. (2014, except experiment 1) conducted several studies with different font manipulations, but they could not detect better performance in the disfluent condition or found even partly worse performance of the disfluent condition. A whole series of studies on the disfluency effect recently published in a special issue (Metacognition and Learning, 2016) could also not show a general disfluency effect. Instead, several moderating effects are discussed like learners characteristics or learners goals. The study of Lehmann et al. (2016) could only show the disfluency effect for learners with higher working memory capacity, indicating that the metacognitive trigger that activates System 2 could only be successful if there are sufficient resources to do so. In another study of Eitel and Kühl (2016) the disfluency effect disappeared when the intention to learn is induced by test expectancy which in any case should lead to the activation of System 2 and hence competes with the metacognitive trigger of disfluency itself.

One explanation of the heterogeneous findings on disfluency we focus on in this paper could be the various manipulation methods used which might have led to different disfluency levels and thus to different ways of information processing (Alter and Oppenheimer 2009). These methods include (among others) font manipulations (e.g. French et al. 2013), blurring (e.g. Yue et al. 2013), poor quality copies (e.g. Eitel et al. 2014), inverted fonts (e.g. Sungkhasettee et al. 2011), adverse figure-ground contrasts (Reber and Schwarz 1999) and combinations of two or more of these manipulations (e.g. Diemand-Yauman et al. 2011). Especially font manipulations are commonly utilized to generate disfluency (Alter and Oppenheimer 2009). But even in this category we have to differentiate between the use of different font sizes (Rhodes and Castel 2008), difficult to read fonts like Monotype Corsiva (French et al. 2013), italicized fonts (Alter et al. 2007) or fonts with different gray contrasts (Alter et al. 2007; Diemand-Yauman et al. 2011; Eitel et al. 2014). Based on this differentiation the question arises whether these several font manipulations generate the same legibility of the used materials—even when very similar methods were used.

The first problem with the mentioned studies is that some used only one manipulation; others combined several at once. For example French et al. (2013) used only the difficult to read font Monotype Corsiva, whereas Diemand-Yauman et al. (2011) and Eitel et al. (2014) utilized three manipulations: different fonts, grey font contrast and poor quality copies. Although the last two studies used very similar methods, they differed in the degree of contrast, for example: the first one chose a grey contrast of 60 % whereas in the latter study a grey contrast of 35 % was used. Hence, it is questionable whether these three studies generate the same disfluency level.

The second problem is the fact that the mentioned studies did not only use various kinds of disfluency manipulations but also used different fluent conditions as a control condition. Alter et al. (2007) as well as Diemand-Yauman et al. (2011) for instance used larger easy to read fonts in comparison to the disfluent font. Yue et al. (2013) or Eitel et al. (2014) on the other hand utilized the same font size for the fluent and disfluent condition. This is important to be considered because Diemand-Yauman et al. (2011) supposed that the neutral effect found by Rhodes and Castel (2008) could be due to their used font sizes (from 18 to 48 pt.) which were larger than the normal standard sizes and thus not disfluent enough. Conversely, it might be possible that smaller fonts used in fluent conditions demonstrate the same neutral effect, because the fluent font was not fluent enough or could even be perceived as slightly disfluent, respectively. In this case some of the control conditions used maybe also induced a more elaborated processing style to some degree.

In conclusion, the heterogeneous effects of disfluency manipulations on performance may be a result of the inconsistency in the disfluent as well as fluent conditions. This view is supported by Diemand-Yauman et al. (2011), who supposed that the influence of disfluency on performance follows a reversed u-shaped curve. In other words: low and very high levels of disfluency show either none, only slightly positive influences or even a negative impact on performance whereas middle levels of disfluency result in the presumed positive effect on learning and performance. While too low levels of disfluency on the one hand may not be perceived as different from fluent text, it will not suffice to trigger deeper processing. Too much disfluency on the other hand may impose too many difficulties and hence hinder learning. Thus, it is necessary to find the optimal level of disfluency that we assume to be on a medium level of legibility.

The present study

The present study investigates an alternative approach to clarify the mixed evidence in disfluency research. Based on the assumption that different disfluency levels have different impacts on learning and learning performance, we created four versions of learning material with increasing disfluency levels (fluent, disfluency level 1, disfluency level 2, disfluency level 3) and subsequently tested participants on their learning outcomes. In accordance to Diemand-Yauman et al. (2011) we presume the following: The connection between legibility of the learning material and learning outcomes shows a reversed u-shaped pattern.

Method study 1

Participants and design

To answer the question whether different levels of text legibility have the expected effects on learning, we compared four experimental groups with respect to their learning outcomes. In this study 67 students from a german university participated for course credits. There were 61 female and 6 male participants ($M = 21.3$ years, $SD = 2.58$). They were randomly assigned to one of the four groups of the independent variable *legibility* (fluency, disfluency- level 1, disfluency- level 2, disfluency- level 3). As dependent variables we measured participants' learning outcomes directly after they finished learning the text. Furthermore we included learners' spatial abilities and their current motivational state as

control variables that could explain for variance of learning performance (Eitel et al. 2014; Moreno 2005; Rheinberg et al. 2001). Due to the fictional story used in the learning material we did not have to assess learners' prior knowledge. Based on the study of Lehmann et al. (2016) we also took into account learners' working memory capacity. Due to their study only learners with higher capacity levels benefited from disfluency, so we controlled for working memory capacity.

Learning material

All materials were printed on sheets of paper. The text-based instructional material was a fictional story about a chocolate factory (270 words) that can be read in about 3–4 min. In this study the text's legibility was manipulated in four different conditions by changing the text's font, size and contrast: Fluent (Times New Roman, black, 12 pt); Disfluency-1 (Haettenschweiler, black, 14 pt); Disfluency-2 (Haettenschweiler, 25 % grey, 14 pt) and Disfluency-3 (Haettenschweiler, 15 % grey, 14 pt). Overall, we varied the font and size from fluent to disfluent, but within the disfluency levels we only varied the contrast.

Measures

Beside a demographic questionnaire asking for age and gender we measured *task related motivation* (differentiated in interest, probability of success, fear of failure and challenge) with the Questionnaire on Current Motivation (QCM; Rheinberg et al. 2001). The Numerical Updating subtest of the WMC test (Oberauer et al. 2000) was used to measure participants' *working memory capacity*. In addition we controlled for *learners' spatial abilities*, measured by the paper folding and card rotation test (Ekstrom et al. 1976).

The paper-based test for *learning outcomes* comprised 11 questions of different formats: multiple-choice ("What was the name of Oliver's father?" Heinrich–Hans–Harald–Horst), open questions where single words or a certain number of aspects had to be named and were valued ("What four kinds of foods does the fabric produce?", "Name the three cities that are mentioned in the text") and open questions where learners had to explain the correct answer ("The delivery point for almonds is located southwest of the fabric. Why is this place impractical from a logistical point of view? Which location would be a better solution?"). One to five points were given for correct answers (one point for each correctly answered aspect) adding up to a potential maximum of 22 points for learning outcomes.

Additionally we measured learners perceived quality of layout ("I perceived the layout of the material as well designed") and legibility ("I perceived the font as easy to read") as a manipulation check. All measures except the working memory test were assessed in a paper-based format.

Procedure

First, participants were randomly assigned to one of the four groups. After filling in the demographic questionnaire, and dealing with the tests for the control variables, participants were asked to begin reading the learning material within a 10 min time frame. After that, participants were asked to rate the learning material concerning the perceived quality of layout and legibility. Finally, all participants simultaneously started filling in the test for learning outcomes without any time restrictions.

Results study 1

First, an analysis of variance (ANOVA) was conducted to check whether age, spatial abilities, working memory capacity and motivation were equally distributed across the four experimental groups. We found significant differences between the groups concerning participants' age ($F(3, 63) = 2.84, p = .05, \eta_p^2 = .12$) and the motivation subscale 'probability of success' ($F(3, 63) = 3.13, p = .03, \eta_p^2 = .10$). Thus, these variables were included in the analysis as covariates. Moreover, as working memory capacity correlated with learning outcomes ($r = .33, p < .01$) we also included it as a covariate. Descriptive data of all variables are reported in Table 1.

Manipulation check

An ANOVA with the two manipulation check items as dependent variables was conducted. Results revealed a significant effect for the materials' perceived quality ($F(3, 63) = 3.00, p = .04, \eta_p^2 = .16$) and the texts' perceived legibility ($F(3, 63) = 35.79, p < .001, \eta_p^2 = .63$). As expected, the polynomial contrasts revealed a linear trend for both variables (layout quality: $MD = 1.16, SE = 0.40, p < .01$; legibility: $MD = 3.36, SE = 0.35, p < .001$): the learners' perception of the texts' layout and legibility declined continuously from the fluency condition to the third disfluency condition as can be seen in Table 1.

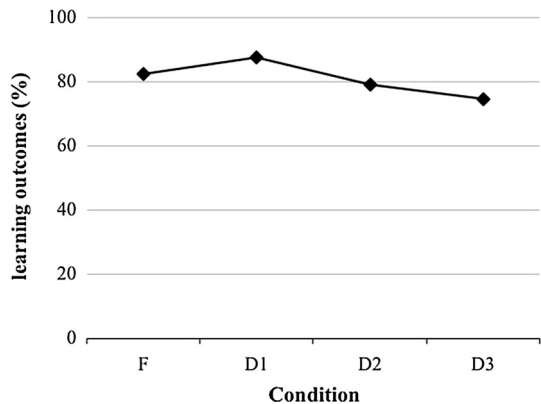
Learning outcomes

A one-factorial ANCOVA with text legibility as independent variable and the learning outcomes scores as dependent variable was conducted, with age, probability of success and working memory capacity as covariates. Results revealed a significant effect for text legibility ($F(3, 54) = 3.50, p = .02, \eta_p^2 = .16$). Contrasts between subgroups showed that the first disfluency group outperformed the third disfluency group ($MD = 13.02$,

Table 1 Means and standard deviations of all variables for all experimental conditions (study 1)

	Conditions			
	Fluency (<i>n</i> = 17)	Disfluency-1 (<i>n</i> = 17)	Disfluency-2 (<i>n</i> = 17)	Disfluency-3 (<i>n</i> = 17)
Perceived quality of layout (1–7)	4.38 (1.15)	3.59 (1.77)	3.06 (2.56)	2.82 (1.85)
Perceived legibility (1–7)	6.44 (0.63)	3.65 (1.94)	2.12 (1.45)	1.94 (1.25)
Learning Outcomes (%)	82.55 (2.75)	87.72 (2.96)	79.23 (2.94)	74.69 (2.92)
Interest (1–7)	3.31 (1.27)	3.67 (0.60)	3.60 (1.19)	3.69 (0.85)
Probability of success (1–7)	5.22 (0.86)	5.99 (0.70)	5.26 (0.81)	5.46 (0.88)
Fear of failure (1–7)	2.74 (0.78)	2.07 (0.98)	2.56 (1.00)	2.92 (1.12)
Challenge (1–7)	4.59 (0.97)	4.07 (0.93)	4.21 (0.57)	4.34 (1.04)
Working memory capacity (1–9)	3.75 (1.00)	3.93 (1.16)	3.40 (1.30)	3.40 (0.91)
Age (years)	21.19 (2.23)	21.29 (3.55)	22.59 (2.37)	20.12 (1.11)

Fig. 1 Effects of different disfluency levels on learning outcomes for study 1 (F = fluency, D1–3 = disfluency level 1–3)



$SE = 4.17$, $p < .01$; see Fig. 1). Furthermore, polynomial contrast analysis revealed no significant quadratic trend ($MD = 4.85$, $SE = 2.89$, $p < .10$), merely on a descriptive level the u-shaped curve is slightly observable (see Fig. 1).

Discussion of study 1 and implications for study 2

Legibility of a text actually seems to influence learning outcomes. We found a significant effect for the text versions depending on their legibility. Interestingly the contrasts did not reveal the “classical disfluency effect”, i.e., a significant difference between the fluent and the first, i.e. the slightly disfluent group, which is comparable to those often used in previous disfluency studies. Generally the learning outcomes were quite good for all learners so probably the material might have been too easy, resulting in less variance. Nevertheless, we have first hints for a u-shaped relation of disfluency and learning outcomes: disfluency first may foster learning, but can hinder learning if it becomes too disturbing, what is also underlined by the significant difference between the first and the third disfluency level. This is even more interesting as learners in fact did not rate the legibility of the last two disfluency levels significantly different. Maybe we should have used a broader scale for the legibility manipulation check or we could have enhanced the actual legibility by using an even less legible font in the strongest disfluency level.

The results of the first study encouraged us to conduct a replication study with additional aspects.

First and most important we decided to vary only one aspect from the fluent to the disfluent conditions, namely only the font and not its size additionally. For the disfluency levels we then also varied only one aspect, namely the grey contrast. To further enhance the variance we also used a more complex, non-fictional learning content. In contrast to the first study we then also needed a measure for prior knowledge.

Second, based on previous studies on disfluency and their heterogeneous results for different levels of learning outcomes, i.e., recall or comprehension or even transfer we decided to differentiate our learning criteria as well. With this we want to identify at which level of processing disfluency actually works and whether the reversal point of the reversed u-shaped relation will differ for the different levels of processing. Especially higher performance levels such as transfer levels might benefit from disfluency, because a more analytical processing style triggered by disfluency should improve understanding of complex contents (Bjork 1994; Eitel et al. 2014). First evidence for that assumption is

provided by Eitel et al. (2014) who found a significant effect of disfluency on transfer performance, but not on retention.

Third, very often the issue of disfluency is also discussed with respect to cognitive load. The underlying assumption is that the effects of text legibility can be linked to the amount of cognitive capacity learners need to invest to process learning material effectively. Recent papers on Cognitive Load Theory (Sweller 2010; Sweller et al. 2011) differentiate between cognitive resources that are related to either task-relevant aspects and therefore are productive (intrinsic) or to task-irrelevant aspects, that are not productive in terms of the goal to be achieved (extraneous). *Intrinsic cognitive load* is caused by the element interactivity of a specific learning task. It is thereby correlated with the amount of elements a learner has to keep in mind and process simultaneously and learners' prior knowledge. Nevertheless, this type of load should not be affected by text legibility, as it does not alter the amount and interrelatedness of the inherent elements of the learning task. *Extraneous cognitive load* on the other hand should be affected by legibility, as it is related to the format of instruction. Poorly designed material requires learners to deal with issues like navigating or searching that do not contribute to learning but nevertheless require cognitive resources. According to Cognitive Load Theory, less legible text can be referred to as extraneous cognitive load and may hinder task related learning processes. It can be assumed that with increasing disfluency the amount of perceived extraneous load should also increase (Eitel et al. 2014). This is contradictory to the assumptions based on research of desirable difficulties. Hence, even more it would be interesting to analyze the effects of text legibility on extraneous cognitive load, especially with regard to the fact, that the cognitive load was not measured in connection with disfluency so far.

In this paper we are additionally interested in those resources the learner allocates to schema acquisition, formerly called *germane cognitive load* (Bannert 2002; Sweller et al. 1998). With an extended view on this concept we assume that due to disfluency as a metacognitive cue to invest more mental effort in processing, learners may in fact invest more of these resources. Moreover, the metacognitive regulation to intensify cognitive processing may also be accompanied by an increase of motivation. We therefore call this aspect of invested resources *engagement*. Like for learning outcomes the effects of increasing disfluency on engagement should also follow a reversed u-shaped form, because there may be a turning point where learners cannot invest additional resources regarding the also increased extraneous load. Generally, it would be interesting to analyze the effects of disfluency on all types of cognitive load and engagement and thus a differentiated measurement could be a fruitful approach. Whether learners do not only invest more mental effort but also more time for learning we additionally took into account learners' time on task.

In the second study we added two more learner characteristics that could influence learners' reaction on the metacognitive cue of disfluency. First, the need for cognition, which was also included in a study by Kühl et al. (2014), because people who score rather high on the need for cognition scale are supposed to invest more effort in learning processing and consequently use rather systematic strategies (Cacioppo and Petty 1982). Second, due to the fact that we now used learning material which was non-fictional learners' prior knowledge had to be taken into account, because it influences the quality and speed of information reception and processing (Kalyuga 2005).

Overall the research question of the second study is similar to the first study: What are the effects of different disfluency levels on learning outcomes, this time differentiated on different levels of processing and additionally on different types of cognitive load, engagement and on time on task.

Method study 2

Participants and design

Following the design of the first study, four experimental groups were compared with respect to their learning outcomes, learning time and subjective ratings of three types of cognitive load. The sample comprised 73 students from a German university with 67 female and 6 male participants who participated in this study for course credits. Their mean age was 21.62 years ($SD = 3.34$). Participants were randomly assigned to one of the four conditions with increasing disfluency (with $n = 18$ per condition except disfluency level 2 with $n = 19$). As dependent variables we assessed participants' learning outcomes (i.e., recall, comprehension and transfer), learning time and cognitive load (i.e., extraneous, intrinsic load and engagement). Moreover we controlled for learners' prior knowledge, their spatial abilities, working memory capacity, motivation and their need for cognition.

Learning material

The learning material was a scientific text about the earth's different time zones (adapted from Schnotz and Bannert 1999) with 1100 words that could be read and studied in about 13 min ($M = 763$ s, $SD = 227.51$). It also included a table with the time differences of the eight cities compared to Greenwich time, which were included in the text and the subsequent test. The text's legibility was manipulated in four different conditions by changing the text's font and contrast: Fluent (Arial, black, 12 pt); Disfluency-1 (Monotype Corsiva, black, 12 pt); Disfluency-2 (Monotype Corsiva, 25 % grey, 12 pt) and Disfluency-3 (Monotype Corsiva, 15 % grey, 12 pt). So overall, we changed the font but not the font size from fluent to disfluent and within the disfluency levels we only varied the contrast. All materials were printed on sheets of paper.

Measures

Measures comprised a demographic questionnaire for age and gender of participants, the *questionnaire for current motivational state* (QCM), a test assessing participants' *working memory capacity* and a test for *spatial abilities*, which were identical to the instruments used in study 1. In this study, participants' need for cognition was also assessed by using a short version of a questionnaire developed by Bless et al. (1994). We again used two items for evaluating the learning material as a manipulation check. All instruments except the working memory test were assessed in a paper-based format.

The paper-based test for *prior knowledge* comprised six tasks on time differences on earth. Four of them were of open format; two of multiple-choice format and learners could use a table with time differences for several cities compared to Greenwich (e.g. Anchorage -10 , Los Angeles -8 , Moscow $+3$, Tokyo $+9$). One open question was for example which time it is in Tokyo if it is 3 pm in Moscow. One point was given for the correct answer 9 pm, another one for the correct approach, i.e. the calculation based on the differences in the given table: $(+3) - (+9) = (-6)$ either in verbal or in number format. In that way all tasks were coded with a maximum of two points for either the correct answer and its specification, or for identifying the two correct answers in the multiple-choice task (two correct answers and two distractor sentences, with a reduction for false responses) or for naming the two relevant aspects in one task (e.g. what is a time-zone (one point) and how many of them are there (one

point)). The internal consistency was quite low with Cronbach's $\alpha = .40$, but as the test addresses very different aspects of the content domain which are not necessarily linked to each other the internal consistency must naturally be low.

The paper-based test for *learning outcomes* consisted of three subtests to measure recall, comprehension and transfer. Two points were given for correct answers in each subtask in the same fashion as in the prior knowledge test. Recall was assessed with four open tasks (i.e., "What is the meridian?") and one multiple-choice task (i.e., "At which position of the sun is it exactly 12 am at any place and at which position of the sun is it exactly 12 pm at any place?"), resulting in a maximum of ten points for recall. The internal consistency was quite low with Cronbach's $\alpha = .50$, but as the test addresses very different details of the text to be remembered the internal consistency must again be naturally low. Comprehension of the consistent concept of time differences on earth was measured with six open questions (i.e., "When it is 3 pm in Frankfurt, what time is it in Anchorage?"). Two points were given for correct answers resulting in a maximum of 12 points for the comprehension test with a very high $\alpha = .93$. The transfer test also consisted of six open questions (i.e., "You are arriving in Tokyo on the 24th of June at 4 pm after a 12 h flight. On what date and time did your flight take off in Greenland?"). Again a maximum of 12 points could be achieved for the transfer score with $\alpha = .69$.

To measure intrinsic, extraneous load and engagement a paper-based questionnaire (Klepsch and Seufert 2012) was used consisting of eight items on a seven point Likert scale ranging from 1 (=absolutely not right) to 7 (=absolutely right). To measure intrinsic load two items were included ((1) "This task requires to process multiple information simultaneously", (2) "This task is very complex") with an internal consistency of $\alpha = .81$. Three items were used to assess extraneous cognitive load ((1) "It is hard to identify the most important content in this material", (2) "The design of the task is inappropriate for learning", (3) "It's difficult to see the connections between the different aspects") with an internal consistency of $\alpha = .73$. Participants rated their engagement with three items ((1) "My goal was to gain deep knowledge of the material", (2) "I invested much effort to not only memorize details but to get an overall understanding", (3) "I strained myself") with an internal consistency of $\alpha = .78$.

Procedure

First, participants were randomly assigned to one of the four groups. After filling in the demographic questionnaire participants were asked to answer the questions of the prior knowledge test and to deal with all the other measures for control variables. All learners started to read the text at the same time and a timer was started and displayed in front. Participants were then asked to note the time when they finished reading the text. There was no time limit. Afterwards, participants had to subjectively rate their cognitive load during reading and learning and to evaluate the text layout and legibility before they were given the test for learning outcomes.

Results study 2

First, an analysis of variance (ANOVA) revealed that the groups differed with respect to learners' spatial abilities but in none of the other control variables, hence spatial abilities were included as a covariate in all subsequent analyses. The other control variables were

included in the analyses as covariates and reported below when they were significantly correlated with the respective dependent measure indicating a co-dependency that should be eliminated. Descriptive data of all control variables are reported in Table 2.

Manipulation check

An ANOVA with the two manipulation check items as dependent variables was conducted. As expected, results revealed a significant effect for the *perceived quality of layout* ($F(3, 69) = 11.68, p < .001, \eta_p^2 = .42$). Polynomial contrasts showed a significant linear trend ($MD = -2.12, SE = .36, p < .001$). The perceived quality was rated higher in the fluent group than in the disfluency condition 2 and 3 ($MD = 2.01, SE = .50, p < .001$; $MD = 2.72, SE = .51, p < .001$). Also, the layout was rated higher in the first disfluency condition than in the second and third disfluency condition ($MD = 1.31, SE = .51, p = .01$; $MD = 1.94, SE = .51, p < .001$).

The conducted ANOVA also showed a significant impact on *perceived text legibility* ($F(3, 69) = 110.62, p < .001, \eta_p^2 = .83$) with a significant linear trend ($MD = -4.13, SE = .24, p < .001$). The fluent condition was rated more legible compared to all other conditions (disfluency 1: $MD = 2.28, SE = .34, p < .001$, disfluency 2: $MD = 5.08, SE = .33, p < .001$, disfluency 3: $MD = 5.22, SE = .34, p < .001$). Legibility in the first disfluency condition was also rated higher than in both other conditions (disfluency 2: $MD = 2.80, SE = .33, p < .001$, disfluency 3: $MD = 2.94, SE = .34, p < .001$).

Learning outcomes

ANCOVAs were conducted to compare recall, comprehension and transfer scores between the four conditions. Means and standard deviations for all dependent measures can be seen in Table 3 (see also Fig. 2).

Table 2 Means and standard deviations of all control variables for all experimental conditions (study 2)

	Conditions			
	Fluency (<i>n</i> = 18)	Disfluency-1 (<i>n</i> = 18)	Disfluency-2 (<i>n</i> = 19)	Disfluency-3 (<i>n</i> = 18)
Perceived quality of layout (1–7)	4.72 (1.81)	3.94 (1.40)	2.63 (1.54)	2.00 (1.37)
Perceived legibility (1–7)	6.56 (.78)	4.28 (1.45)	1.47 (0.70)	1.33 (.97)
Prior knowledge (%)	56.71 (16.30)	49.54 (19.49)	49.12 (20.01)	57.87 (19.06)
Interest (1–7)	3.46 (1.32)	3.27 (1.16)	2.94 (1.02)	3.73 (.75)
Probability of success (1–7)	5.10 (1.02)	5.15 (1.07)	5.38 (1.00)	5.38 (1.01)
Fear of failure (1–7)	2.87 (1.28)	2.53 (1.18)	2.44 (1.06)	2.74 (1.19)
Challenge (1–7)	4.29 (1.14)	4.36 (0.86)	4.12 (0.74)	4.40 (1.03)
Working memory (1–9)	3.59 (1.23)	3.69 (1.08)	3.78 (1.11)	3.50 (1.10)
Need for cognition (1–7)	4.40 (0.95)	4.57 (0.85)	4.35 (0.74)	4.42 (0.69)
Spatial ability (%)	73.53 (11.97)	73.29 (12.82)	62.55 (14.30)	65.93 (17.36)
Age (years)	21.06 (2.49)	21.83 (4.72)	21.32 (2.06)	22.28 (3.66)

The ANCOVA for *recall* revealed a significant difference ($F(3, 67) = 2.75, p < .05, n_p^2 = .11$). Covariates were spatial abilities due to the group differences and in addition learners' prior knowledge ($r = .38, p = .001$). We also found a significant quadratic trend ($MD = -8.99, SE = 3.94, p < .05$) as can be seen in Fig. 2 (left side) and the second disfluency group outperformed the third disfluency group significantly ($MD = 15.61, SE = 5.50, p < .01$).

For *comprehension* performance we also conducted an ANCOVA [covariates were spatial abilities and prior knowledge ($r = .50, p < .001$)] which revealed no main effect ($F < 1$, n.s.) and no quadratic trend ($MD = -3.13, SE = 7.30, p = .67$) or any other significant contrasts between the groups (see Fig. 2 in the middle).

However, results of the ANCOVA for *transfer* [covariates were spatial abilities, prior knowledge ($r = .40, p < .001$) and need for cognition ($r = .26, p < .05$)] revealed no significant effect ($F(3, 66) = 1.61, p = .19, n_p^2 = .07$) but a significant quadratic trend ($MD = -15.38, SE = 7.17, p < .05$) (see Fig. 2, right side). There were no other significant contrasts between the groups.

Cognitive load

ANCOVAs were conducted to answer the question whether the different learning material conditions show any impact on intrinsic, extraneous load and engagement. As expected, the ANCOVA [with spatial abilities and prior knowledge ($r = -.27, p < .05$) as covariates] showed no impact on *intrinsic cognitive load* ($F(3, 67) = 1.38, p = .26, n_p^2 = .06$) and no significant quadratic trend ($MD = .58, SE = .31, p = .06$). On a descriptive level the curve looks nevertheless u-shaped and especially the third disfluency group reported high intrinsic load scores (see Fig. 3 on the left side).

Results of the ANCOVA for *extraneous cognitive load* [covariates were spatial abilities, need for cognition ($r = -.30, p = .01$) and probability of success ($r = -.25, p < .05$)] revealed a significant difference between the four conditions ($F(3, 66) = 5.58, p < .01, n_p^2 = .21$). According to the hypotheses, polynomial contrasts showed a linear trend ($MD = .83, SE = .28, p < .01$) and an unexpected quadratic trend ($MD = 0.67, SE = .27, p = .01$). Contrasts between the groups indicated significant higher extraneous load ratings of the third disfluency group compared to all the others (fluent: $MD = -1.25, SE = .39, p < .01$, disfluency 1: $MD = -1.29, SE = .39, p = .001$; disfluency 2: $MD = -1.30, SE = .37, p = .001$; see Fig. 3 on the left side).

Table 3 Means and standard errors of all dependent variables for all experimental conditions (study 2)

	Conditions			
	Fluency (<i>n</i> = 18)	Disfluency-1 (<i>n</i> = 18)	Disfluency-2 (<i>n</i> = 19)	Disfluency-3 (<i>n</i> = 18)
Recall (%)	63.93 (3.94)	66.31 (3.96)	73.83 (3.88)	58.21 (3.94)
Comprehension (%)	63.57 (7.30)	76.96 (7.33)	66.90 (7.19)	74.04 (7.29)
Transfer (%)	44.61 (7.16)	57.55 (7.20)	57.89 (7.04)	40.08 (7.14)
Learning time (s)	639.80 (52.93)	777.32 (52.85)	791.01 (52.07)	842.43 (52.43)
Intrinsic load (1–7)	4.70 (0.31)	4.30 (0.31)	4.18 (0.30)	4.95 (0.31)
Extraneous load (1–7)	3.12 (0.27)	3.08 (0.27)	3.07 (0.27)	4.37 (0.27)
Engagement (1–7)	5.15 (0.20)	5.71 (0.20)	5.76 (0.20)	5.62 (0.20)

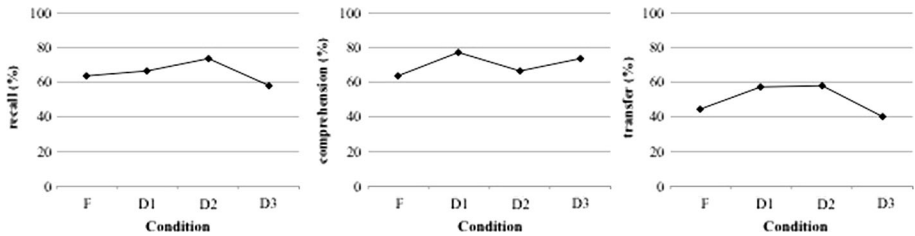


Fig. 2 Effects of different disfluency levels (on recall, comprehension and transfer performance for study 2 (F = fluency, D1-3 = disfluency level 1-3)

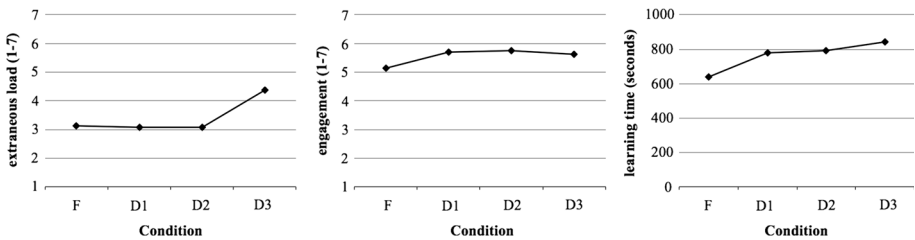


Fig. 3 Effects of different disfluency levels on extraneous load, engagement and time on task for study 2 (F = fluency, D1-3 = disfluency level 1-3)

The ANCOVA for *engagement* [covariates were spatial abilities, challenge ($r = .36$, $p < .01$) and prior knowledge ($r = .35$, $p < .01$)] revealed no significant impact of text legibility ($F(3, 66) = 1.97$, $p = .13$, $\eta_p^2 = .08$) and also the polynomial contrasts showed no significant quadratic trend ($MD = -.35$, $SE = .20$, $p = .08$) and also no linear trend ($MD = .33$, $SE = .20$, $p = .11$). The figure (Fig. 3 in the middle) shows nevertheless that the fluent group reports less engagement than the others; compared to the first and the second disfluency level this contrast is even significant ($MD = -.56$, $SE = .28$, $p < .05$; $MD = -.60$, $SE = .29$, $p < .05$).

Time on task

An ANCOVA with text legibility as independent variable, learning time as dependent variable and spatial abilities as covariate was conducted. Results revealed no significant effect but a remarkable effect size ($F(3, 68) = 2.64$, $p = .057$, $\eta_p^2 = .10$). The time learners needed to read and work through the text continuously increased from the fluent condition to the third disfluent condition. Polynomial contrasts showed a significant linear trend ($MD = 138.99$, $SE = 53.95$, $p = .01$) as it is shown in Fig. 3 on the right side. Contrasts between the groups revealed a significant difference between the fluency condition and the second and third disfluency condition ($MD = -151.21$, $SE = 75.63$, $p = .05$; $MD = -202.64$, $SE = 75.13$, $p < .01$).

General discussion

In two experimental studies we analyzed whether different levels of disfluency affect learning. The idea to induce a learning difficulty and hope for positive effects on learning (Alter et al. 2007) is quite amazing and contradicts other didactic approaches like the

Cognitive Load Theory (Sweller et al. 1998). Cognitive Load Theory proposes that extraneous aspects of learning need to be reduced such that learners' working memory capacity will not be overloaded. The freed up resources can then be used for generative processing (Mayer 2014). However, there has been empirical evidence for positive effects of disfluency and it is argued that due to a metacognitive regulation process learners evaluate disfluent materials as more difficult and therefore activate deeper learning processes. Beside positive effects of disfluency on learning there is also vast evidence for none or negative effects of disfluency (Kühl and Eitel 2016). Facing this heterogeneous pattern of effects we especially addressed the question whether the actual level of disfluency could explain for either positive or negative effects on learning. Whereas slight levels of disfluency might activate increased engagement, too much disfluency may only hamper processing.

In our first study we found significant differences between the experimental groups, but only on a descriptive level a reversed u-shaped pattern of disfluency effects on learning outcomes. Surprisingly—or not—the direct comparison of the disfluency levels with a fluent condition did not reveal significant improvement. This is in line with previous studies where also no (significant) disfluency effects could be shown (Kühl and Eitel 2016). Nevertheless, on a descriptive level, learners with the slightly disfluent text outperformed the fluent text group. Moreover, there is first evidence that too much disfluency is in fact detrimental for learning. Both results, the reversed u-shaped pattern and the breaking point, encouraged us to elaborate our analysis.

In the second study we finally found a significant reversed u-shaped pattern for learning outcomes for recall and transfer performance but not for comprehension. The result pattern for comprehension is quite peculiar because we can observe a u-shaped curve from fluent to the second level of disfluency, but the third disfluency group shows unexpected high scores of comprehension. We have no explanation based on all the other results for this group. Their increased engagement and learning time and especially their high extraneous and intrinsic load ratings would suggest the same learning impairment as for recall and transfer. Maybe these learners set their goal on especially this aspect of learning, i.e. they concentrate on understanding the mechanism of time differences on earth without taking into account too much details or transfer problems. Hence, they allocated their resources especially on this learning aspect. In line with the suggestions of Kalyuga and Singh (2015) the role of learners goals within complex learning situations that might even change during the learning phase should be discussed more thoroughly to understand changing or differing patterns of resource allocation.

Despite the inconsistent results for comprehension, the other result patterns fit quite well into the overall picture. The extraneous load is lowest at a medium level of disfluency and thus the extraneous load curve contrasts the other curves, i.e., the curve is u-shaped. So when there is too much extraneous load, as it seems to be the case for the last disfluency level, learners cannot succeed. To exactly detect the breaking point where disfluency is too much seems not possible concerning the inconsistent results for the different measures. Only the last level of disfluency is definitely too illegible. The pattern of engagement is almost parallel to the reversed u-shaped pattern of recall and transfer. It seems that for transfer tasks learners' engagement in germane processes depends on the amount of disfluency. Only on the highest level of disfluency the increased engagement does no longer pay off. The second indicator for an intensified learning process is time on task. In both studies, we deliberately set no time limits so that time on task was under learner control and hence a part of the metacognitive regulation process. We found a linear trend with increased time investment for increasing disfluency. Learners in fact spent more time with

increasing disfluency but when the extraneous load was too high this investment doesn't pay off either. Hence, learning time is a necessary but not a sufficient condition for increased performance.

From a theoretical point of view one can ask whether disfluency can really affect learning *per se*. In our view it seems more plausible that we should model the total amount of cognitive resources required in a special learning situation, taking into account the given capacity of each learner and to decide whether available resources, task difficulty and disfluency are balanced in a way that the metacognitive cue in fact can be fruitful. This argument is in line with the study of Lehman et al. (2016) where only learners with high levels of working memory capacity succeeded to use disfluency effectively for learning. They argue that only these learners are able to deal with the metacognitive trigger imposed by disfluency because they have sufficient resources to actually engage in deeper learning processes.

One important challenge was to develop learning materials with illegibility levels that not only objectively differ due to well-known layout criteria but also subjectively differ. This prerequisite could be reached in both of our studies. But it's quite interesting that, despite the ratings of the last two disfluency levels were almost similarly low, these two levels produced quite different effects on several dependent measures. It seems that the subjective and conscious rating of legibility cannot directly be translated to predict actual learning processes. A similar result could be found by Sungkhasettee et al. (2011), who did not detect any differences in learners' judgement of future memory performance, which they had to make after every word, despite better recall of disfluent words. Therefore, the activation of a deeper processing system seems not to depend on such conscious ratings but must be triggered without conscious decisions. To learn more about these metacognitive processes it seems necessary to invest more in process measures like thinking aloud or eye movement analyses. Nevertheless, the differentiation of the different aspects of cognitive load and engagement gives some first hints at hindering or activating aspects of learning. The scales produced the theoretically assumed patterns and also have satisfyingly high reliability scores. Thus, they should be used in further studies to learn more about their validity. It is also worth to compare the measure we used to the differentiated measure of cognitive load by Leppink et al. (2013). Generally we are aware that measuring cognitive load subjectively is a methodological problem. Not only the learners' individual understanding of the items, also their often insufficient awareness of their actual load level and especially the awareness for specific aspects of cognitive load has to be considered as a limitation. Additional measures of cognitive load like physiological measures or dual task approaches could be a fruitful supplement.

From a methodological point of view we have to state that the samples of both studies were not representative, because the participants were all psychology students with a high female proportion. Therefore generalization is limited and the studies need to be replicated. Moreover the group sizes were relatively small, so the test power was not optimal and might have contributed to not achieving significance in some cases. Nevertheless, the effect sizes were promising so a replication seems worth it. Another critical methodological aspect is the choice of our dependent learning outcomes measure immediately after learning. Bjork (1994) indicated that desirable difficulty enhances particularly long-term performance whereas normal learning sessions produce better short-term performances. So Hogan and Kintsch (1971) could find that the desirable difficulty conditions showed better performance only after a 48-hour delay whereas the control condition produced better results immediately at the end of the experimental session. Therefore,

future studies should include a follow-up measuring of performance to determine possible long-term effects of disfluency.

For future research a more systematical approach of disfluency or step-by-step manipulation changes are needed. Also a clear definition of fluency might be helpful for operationalizing the control condition. With this we might overcome the current situation with very inconsistent findings of disfluency due to the variety of disfluency as well as fluency manipulations utilized.

Overall we have to better understand the perceptual, motivational, cognitive and metacognitive processes going along with disfluency. One approach could be to analyze whether disfluency may help to overcome metacognitive judgment mistakes like overestimating the own performance like Mengelkamp et al. (2015). They additionally use eye movements to “see” the effects of disfluency as a metacognitive trigger and its possible effects on adapted reading behavior. Another approach from Eitel and Köhl (2016) is to analyze effects of test expectancy as a possible moderator of disfluency effects: when learners are triggered to engage more deeply by being aware of a possible test after learning this might compensate the trigger imposed by disfluency. On both cases we can learn more about the actual metacognitive processes, the monitoring as well as possible regulatory decisions during learning with disfluent texts.

To learn more about the cognitive affordances of disfluent texts, studies on aptitude treatment interactions can also be fruitful. They might indicate which learner characteristic could either enhance or compensate disfluency. Besides the above-mentioned study of Lehmann et al. (2016), which revealed an aptitude treatment interaction with learners’ working memory capacity, also studies with prior knowledge as moderator seem promising. Like learners with high working memory capacity also learners with higher levels of prior knowledge could be especially able to use the disfluency trigger effectively. Their existing knowledge could ease deeper processing and elaboration for which prior knowledge is crucial. However, prior knowledge should not be too high so that learners actually need a trigger for deeper processing (Seufert 2003). Besides learner traits also states like motivation should be crucial for disfluency and ought to be taken into account (Moreno 2005)—either by manipulating the current motivational state or by measuring current motivation during the learning process, and especially before and after the disfluency trigger is being set. With this we could learn whether the metacognitive decision to engage more deeply is also related to motivational changes and whether it is a cause or a reaction of the learners’ motivation. In addition, depending on the education level of the study sample, prerequisites like verbal abilities or reading competencies should be taken into account.

A practical implication based on our studies is difficult to state. We cannot recommend disfluency in general because none of our disfluent groups significantly outperformed the fluent group. With the background of moderating effects like in the study of Lehmann et al. (2016) we could state that it depends on whether learners have sufficient cognitive resources to really engage more deeply. Nevertheless there seem to be other fruitful approaches like simply telling students that there will be a test after learning (Eitel and Köhl 2016), which seems to have the same awakening effects. Generally, the interplay between learners’ resources, their goal setting, their metacognitive and at last also motivational regulation are an interesting field of research which surely can be stimulated by disfluency studies.

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